

Enteric Nervous System Dysfunction:

Manifestations – Mediators – Mitigators



NICOLE MEIER, ND

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Nicole Meier, ND, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

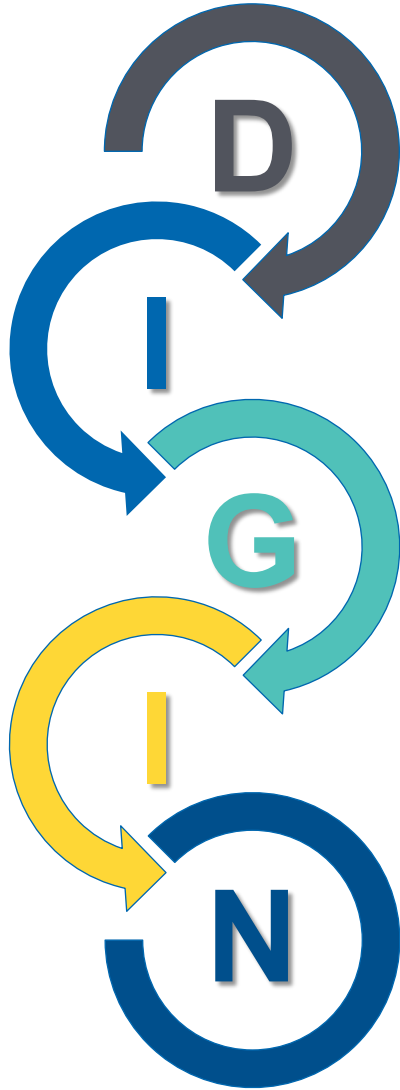
1. Explain the normal physiology of the enteric nervous system and list the other components of the gut-brain axis.
2. Recall common clinical findings, antecedents, triggers, and mediators of gut-brain axis dysfunction.
3. Explain the primacy of acute and chronic stress as mediators of gut-brain axis dysfunction.
4. Explain how gut-brain axis dysfunction may serve as a mediator of extraintestinal disease and/or dysfunction.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Elicit and identify antecedents, triggers, mediators, and clinical findings of gut-brain axis dysfunction and document them on the patient's Timeline and Matrix.
2. Identify intestinal and extraintestinal conditions that may be mediated by gut-brain axis dysfunction.

The DIGIN Mnemonic of Gut Dysfunctions



Digestion/Absorption

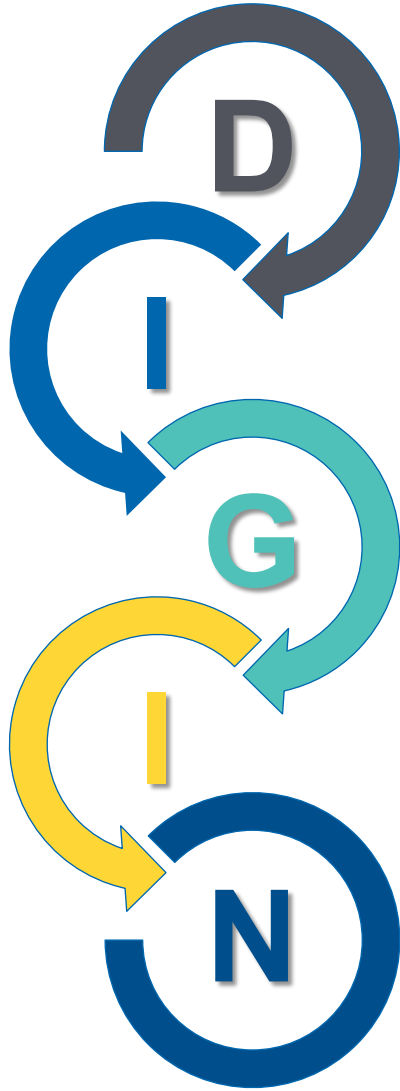
Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System

The DIGIN Mnemonic of Gut Dysfunctions



Nervous System

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Clinical Focus

- Physiology and features of the enteric nervous system and gut-brain axis.
- The enteroendocrine system and its clinical manifestations.



Components of the Nervous System

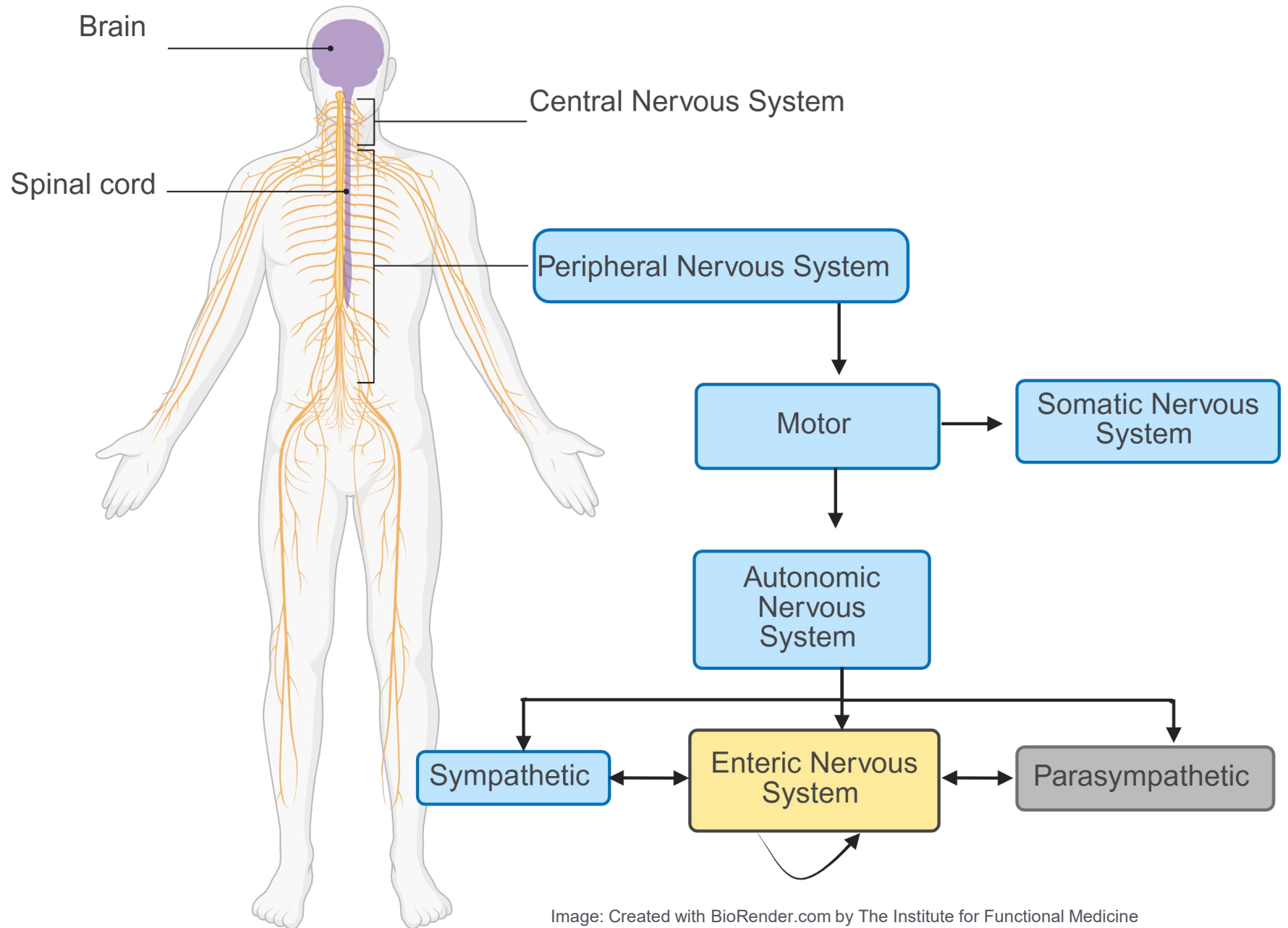


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Long Description: Components of the Nervous System

- The enteric nervous system is one of three branches of the autonomic nervous system. Unlike the parasympathetic and sympathetic branches, it has both motor and sensory functions. The enteric nervous system uses sensory input to coordinate motor function intrinsically.

Gastrointestinal Components of the Nervous System

Gut-Brain Axis (GBA): The coordinated response between the ENS, ANS, and HPA axis via neuro-immuno-endocrine mediators.

- **Enteric Nervous System (ENS):** The axonal network providing a direct connection between the CNS and entire length of the GI tract.
 - Transmits sensory signals bidirectionally between the GI tract/gut microbiome and the CNS.
 - The ENS modulates CNS function; the CNS modulates various GI functions.
- **Vagus Nerve:** Tenth cranial nerve that conveys sensory information from the GI tract (and other organs) to the brain and modulates motor functions.
 - Innervates esophagus, stomach, and small intestine.
 - Communicates between the CNS and ENS to influence GI functions and feeding behavior.

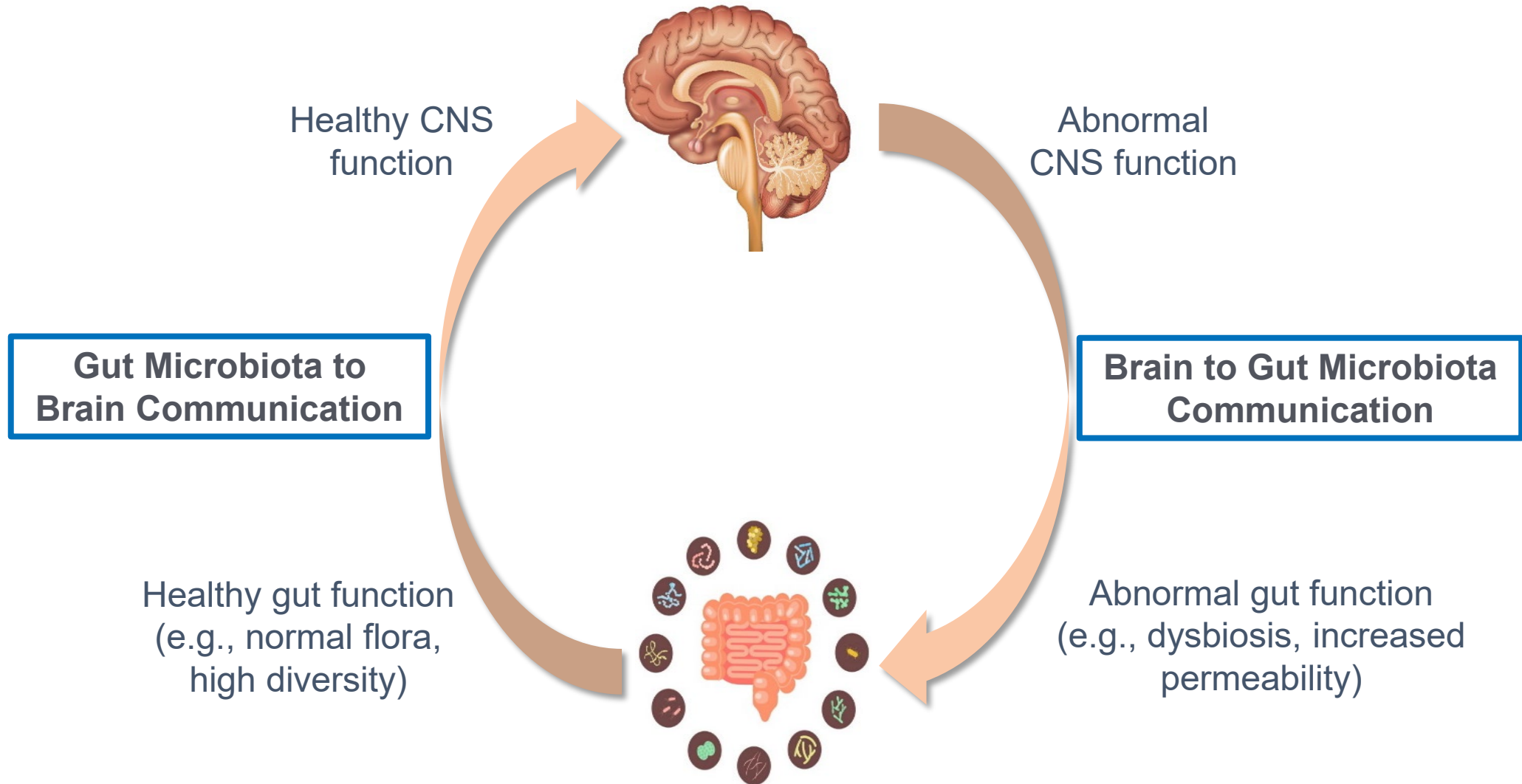
References

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- Chaudhry TS, Senapati SG, Gadam S, et al. The impact of microbiota on the gut-brain axis: examining the complex interplay and implications. *J Clin Med*. 2023;12(16):5231. doi:10.3390/jcm12165231
- Fleming MA 2nd, Ehsan L, Moore SR, Levin DE. The enteric nervous system and its emerging role as a therapeutic target. *Gastroenterol Res Pract*. 2020;2020:8024171. doi:10.1155/2020/8024171
- Liwinski T, Lang UE, Brühl AB, Schneider E. Exploring the therapeutic potential of gamma-aminobutyric acid in stress and depressive disorders through the gut-brain axis. *Biomedicines*. 2023;11(12):3128. doi:10.3390/biomedicines11123128
- Lobo B, Tramullas M, Finger BC, et al. The stressed gut: region-specific immune and neuroplasticity changes in response to chronic psychosocial stress. *J Neurogastroenterol Motil*. 2023;29(1):72-84. doi:10.5056/jnm22009
- Richards P, Thornberry NA, Pinto S. The gut-brain axis: identifying new therapeutic approaches for type 2 diabetes, obesity, and related disorders. *Mol Metab*. 2021;46:101175. doi:10.1016/j.molmet.2021.101175
- Wang SZ, Yu YJ, Adeli K. Role of gut microbiota in neuroendocrine regulation of carbohydrate and lipid metabolism via the microbiota-gut-brain-liver axis. *Microorganisms*. 2020;8(4):527. doi:10.3390/microorganisms8040527

Gut-Brain Axis

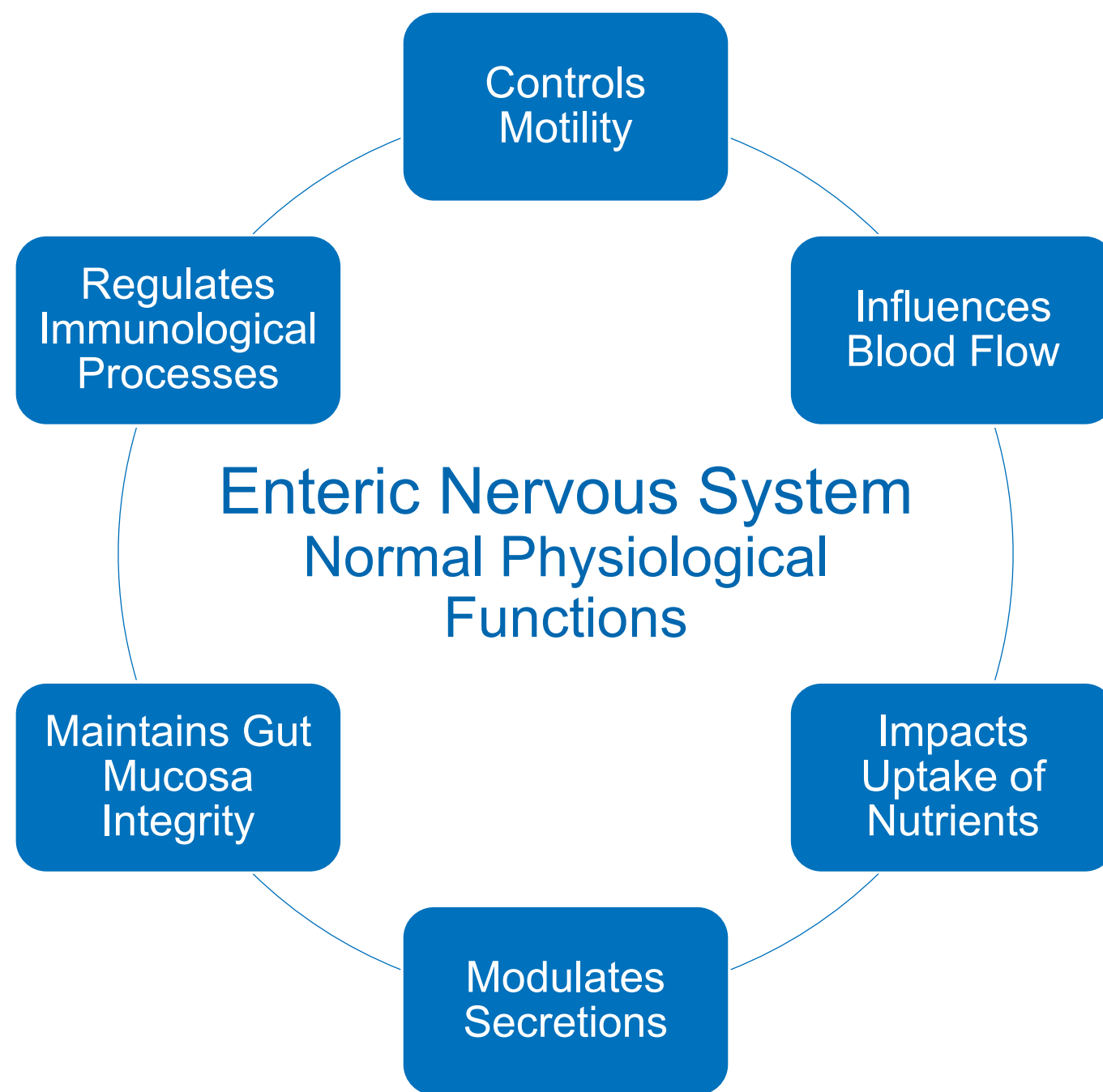
Affects Behavior, Cognition, Emotion, Inflammatory Cells, and Mediators



- Al-Asmakh M, Anuar F, Zadjali F, Rafter J, Pettersson S. Gut microbial communities modulating brain development and function. *Gut microbes*. 2012;3(40):366-373. doi:10.4161/gmic.21287
- Cryan JF, Dinan TG. Mind-altering microorganisms: the impact of the gut microbiota on brain and behaviour. *Nat Rev Neurosci*. 2012;13(10):701-712. doi:10.1038/nrn3346
- Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Gut-Brain Axis Affects Behavior, Cognition, Emotion, Inflammatory Cells, and Mediators

- A bidirectional relationship exists between the gut microbiota and the central nervous system. Healthy gut function supports healthy C N S function, while C N S dysfunction can lead to abnormal gut function.



Enteric Nervous System

Normal Physiological Functions

- **Functions** independently of the central nervous system (CNS) due to autonomous myenteric plexi.
 - Vagus nerve provides a direct connection between GI tract and CNS
- **Controls** motility, blood flow, uptake of nutrients, and secretion.
 - Non-propulsive mixing or segmentation
 - Slow orthograde propulsion via migrating myoelectric complex (MMC)
 - Retropulsion of noxious substances
 - Modification of nutrient handling and changing of local blood flow

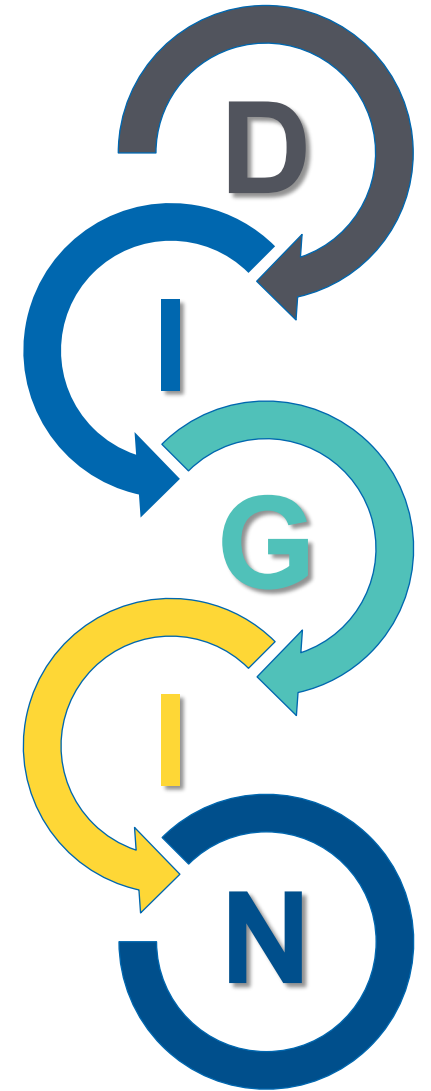


Enteric Nervous System

Normal Physiological Functions

- **Maintains** the integrity of the gut mucosa and regulates its permeability and turnover, thus playing a major role in clinical indicators associated with barrier dysfunction.
- **Influenced** directly by gut bacteria – a bidirectional effect that may contribute to afferent signaling to the brain via the vagus nerve.
 - Modulates the function and the anatomy of the ENS through bacterial metabolites, cytokines, and neurotransmitters such as 5-HT.
- **Regulates** immunological and inflammatory processes in the gut.

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS and Patrick Hanaway, MD, IFM.



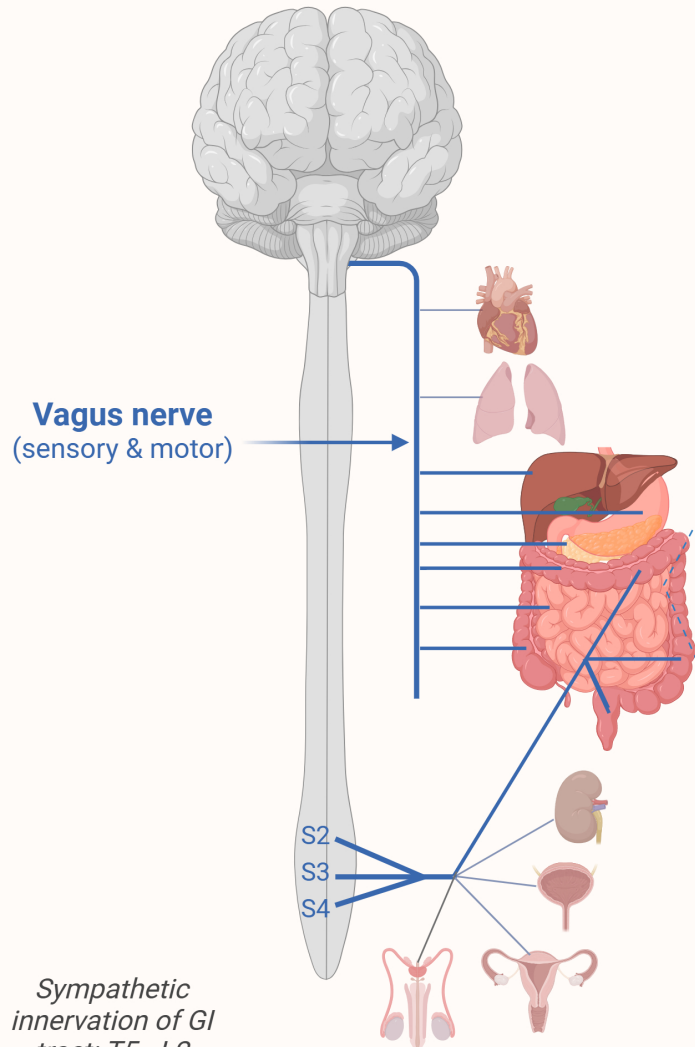
References

ENS: NORMAL PHYSIOLOGICAL FUNCTIONS

- De Vadder F, Grasset E, Mannerås Holm L, et al. Gut microbiota regulates maturation of the adult enteric nervous system via enteric serotonin networks. *Proc Natl Acad Sci U S A*. 2018;115(25):6458-6463. doi:10.1073/pnas.1720017115
- Dicks LMT. Our mental health is determined by an intrinsic interplay between the central nervous system, enteric nerves, and gut microbiota. *Int J Mol Sci*. 2023;25(1):38. doi:10.3390/ijms25010038
- Dicks LMT. Gut bacteria and neurotransmitters. *Microorganisms*. 2022;10(9):1838. doi:10.3390/microorganisms10091838
- Fleming MA 2nd, Ehsan L, Moore SR, Levin DE. The enteric nervous system and its emerging role as a therapeutic target. *Gastroenterol Res Pract*. 2020;2020:8024171. doi:10.1155/2020/8024171
- Linan-Rico A, Ochoa-Cortes F, Schneider R, Christofi FL. Mini-review: enteric glial cell reactions to inflammation and potential therapeutic implications for GI diseases, motility disorders, and abdominal pain. *Neurosci Lett*. 2023;812:137395. doi:10.1016/j.neulet.2023.137395
- Richards P, Thornberry NA, Pinto S. The gut-brain axis: identifying new therapeutic approaches for type 2 diabetes, obesity, and related disorders. *Mol Metab*. 2021;46:101175. doi:10.1016/j.molmet.2021.101175

Mechanisms of Physiological Functions Controlled by the CNS

Parasympathetic GI Innervation (Craniosacral)



*Sympathetic
innervation of GI
tract: T5 - L2*

Parasympathetic & Enteric Nervous System Innervation of GI Tract

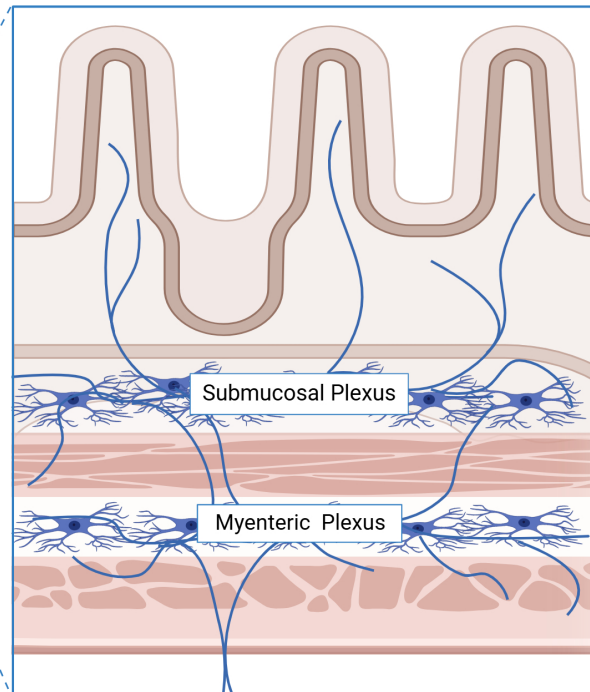


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Enteric Nervous System ("Second Brain")

- Web of sensory, motor, & interneurons embedded in bowel wall
- From lower 1/3 of esophagus to rectum
- Two key layers (ganglia): myenteric & submucosal plexuses
- ENS & CNS communication is bidirectional
- **Can function independently of CNS**

Functions

- Motility - peristalsis/migrating motor complexes
- Secretory - digestive enzymes, fluids
- Blood flow - local regulation, facilitating nutrient absorption
- Sensory - stretch, pH, chemical stimuli
- Neuroendocrine - release of gut hormones
- Neurotransmitters - secreted by ENS; >30 identified, most identical to CNS
- Immune - Development, modulation of intestinal immunity
- Permeability - glial cells help modulate epithelial cells & gut barrier integrity

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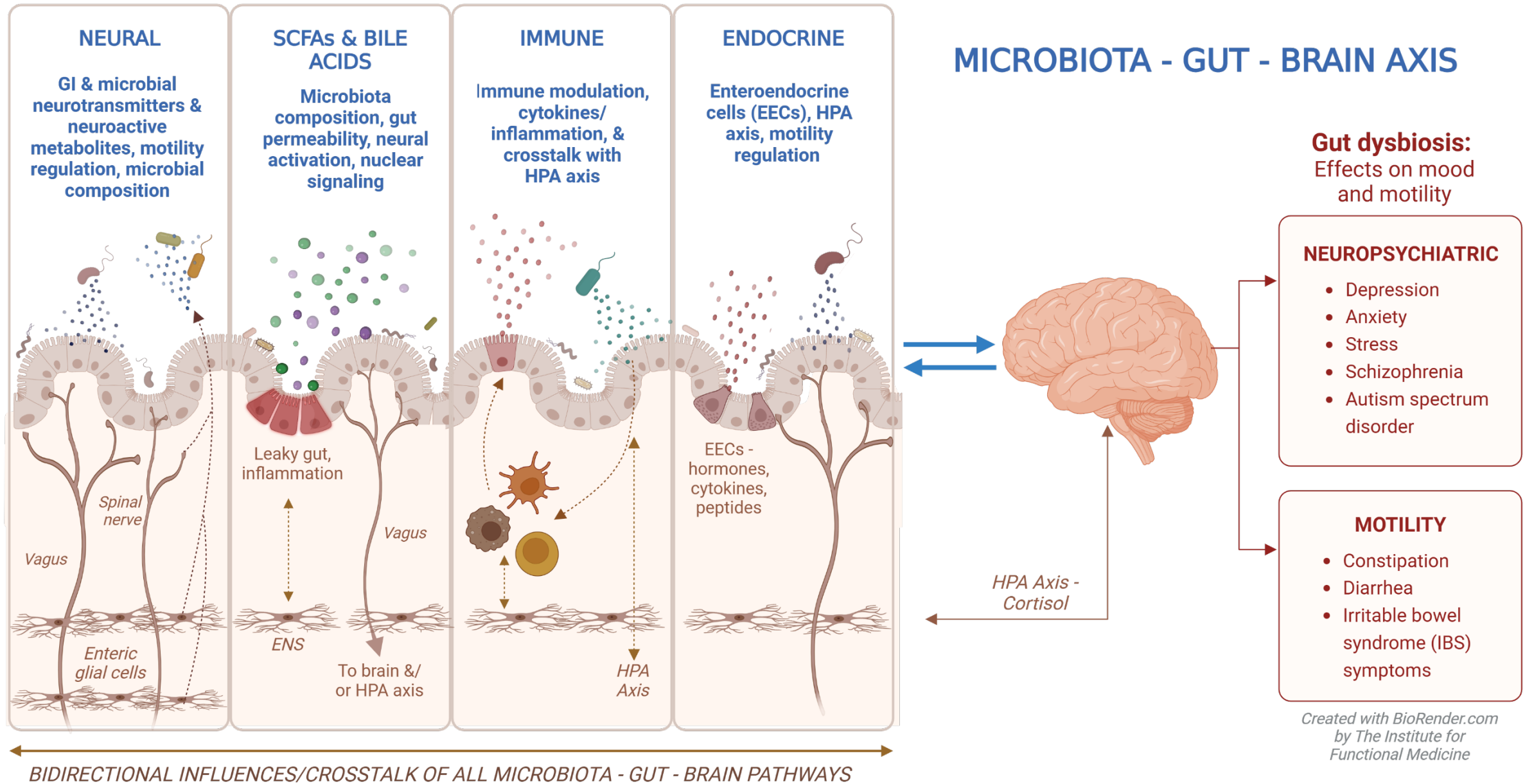
Long Description: Mechanisms of Physiological Functions Controlled by the CNS

- The enteric nervous system is a web of sensory, motor, & interneurons embedded in the bowel wall extending from the lower third of the esophagus to the rectum. The two key layers are the myenteric and submucosal plexuses, which receive sensory inputs from the P N S via the vagus nerve. The myenteric plexus is responsible for motility and contains the migrating motor complexes, while the submucosal plexus governs secretion of digestive enzymes and absorption of nutrients. While the enteric and central nervous system communication is bidirectional, the enteric nervous system can function independently of the C N S.

References

PARASYMPATHETIC AND ENTERIC NERVOUS SYSTEMS

- Chanpong A, Borrelli O, Thapar N. Recent advances in understanding the roles of the enteric nervous system. *Fac Rev.* 2022;11:7. doi:10.12703/r/11-7
- Hansen MB. The enteric nervous system I: organisation and classification. *Pharmacol Toxicol.* 2003;92(3):105-113. doi:10.1034/j.1600-0773.2003.t01-1-920301.x
- McCorry LK. Physiology of the autonomic nervous system. *Am J Pharm Educ.* 2007;71(4):78. doi:10.5688/aj710478
- Nezami BG, Srinivasan S. Enteric nervous system in the small intestine: pathophysiology and clinical implications. *Curr Gastroenterol Rep.* 2010;12(5):358-365. doi:10.1007/s11894-010-0129-9
- Spencer NJ, Hu H. Enteric nervous system: sensory transduction, neural circuits and gastrointestinal motility. *Nat Rev Gastroenterol Hepatol.* 2020;17(6):338-351. doi:10.1038/s41575-020-0271-2



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Long Description: Microbiota Gut Brain Axis

- Bidirectional influences exist between the C N S, the gut, and its microbiota. H P A axis dysregulation can alter immune function, increase intestinal permeability, and potentiate inflammation. Gut dysbiosis is implicated in dysmotility and neuropsychiatric conditions.

References

MICROBIOTA-GUT-BRAIN AXIS: PATHWAYS, MOOD, AND MOTILITY

- Góralczyk-Bińkowska A, Szmaida-Krygier D, Kozłowska E. The microbiota-gut-brain axis in psychiatric disorders. *Int J Mol Sci*. 2022;23(19):11245. doi:10.3390/ijms231911245
- Margolis KG, Cryan JF, Mayer EA. The microbiota-gut-brain axis: from motility to mood. *Gastroenterology*. 2021;160(5):1486-1501. doi:10.1053/j.gastro.2020.10.066
- Mulak A. Bile acids as key modulators of the brain-gut-microbiota axis in Alzheimer's disease. *J Alzheimers Dis*. 2021;84(2):461-477. doi:10.3233/JAD-210608
- Waclawiková B, Codutti A, Alim K, El Aidy S. Gut microbiota-motility interregulation: insights from in vivo, ex vivo and in silico studies. *Gut Microbes*. 2022;14(1):1997296. doi:10.1080/19490976.2021.1997296
- Zhu S, Jiang Y, Xu K, et al. The progress of gut microbiome research related to brain disorders. *J Neuroinflammation*. 2020;17(1):25. doi:10.1186/s12974-020-1705-z

Short Chain Fatty Acids (SCFAs)

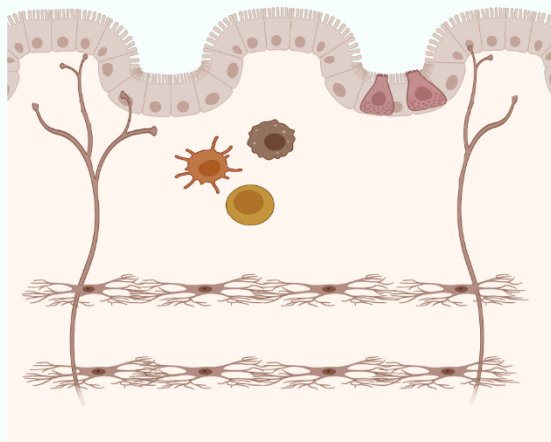
- From microbial fermentation of fibers
- 95% acetate, propionate, butyrate



Stimulation of free fatty acid receptors on multiple cell types:

- Epithelial cells
- Immune cells
- Enterochromaffin cells (ECCs)
- Nerve cells (vagus, spinal, ENS)

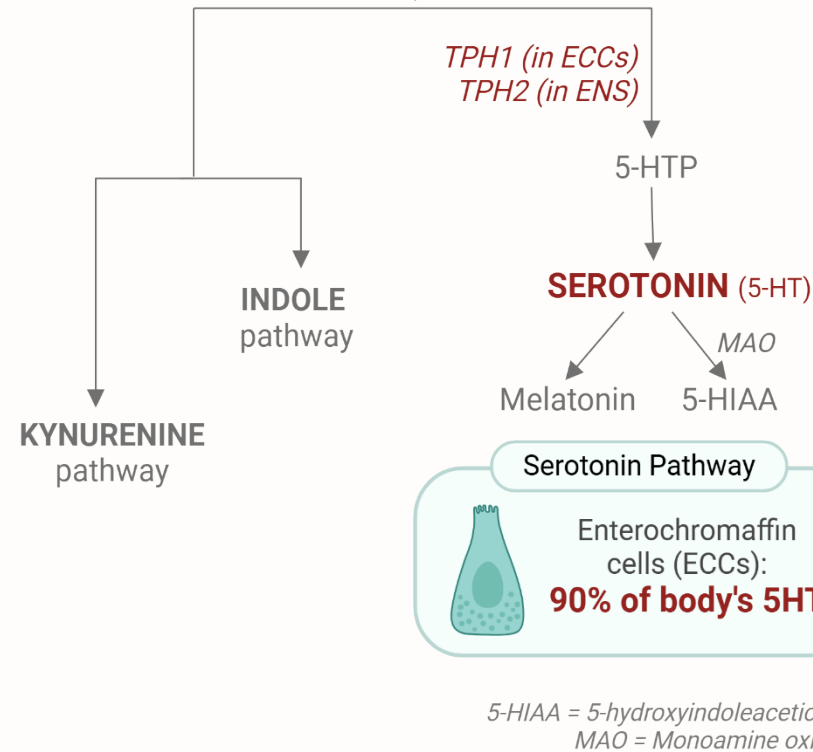
- Regulation of:
- Motility
 - Secretion
 - Gut-brain signaling



SCFAs promote tryptophan hydroxylase 1 (TPH1) transcription in ECCs

Tryptophan Metabolism in the Gut

TRYPTOPHAN



GUT SEROTONIN

- ENS:
Modulates motility, neuronal development, differentiation
- VAGAL/SPINAL SIGNALS TO CNS:
To brain/emotion modulating networks
- ENTEROCYTES:
To bloodstream, platelets
- BIDIRECTIONAL AUTONOMIC NERVOUS SYSEM MODULATION:
Gut function and enteric glial cells
- Influences on mood, gut motility**

SHORT CHAIN FATTY ACIDS, SEROTONIN, AND THE MICROBIOTA-GUT-BRAIN AXIS

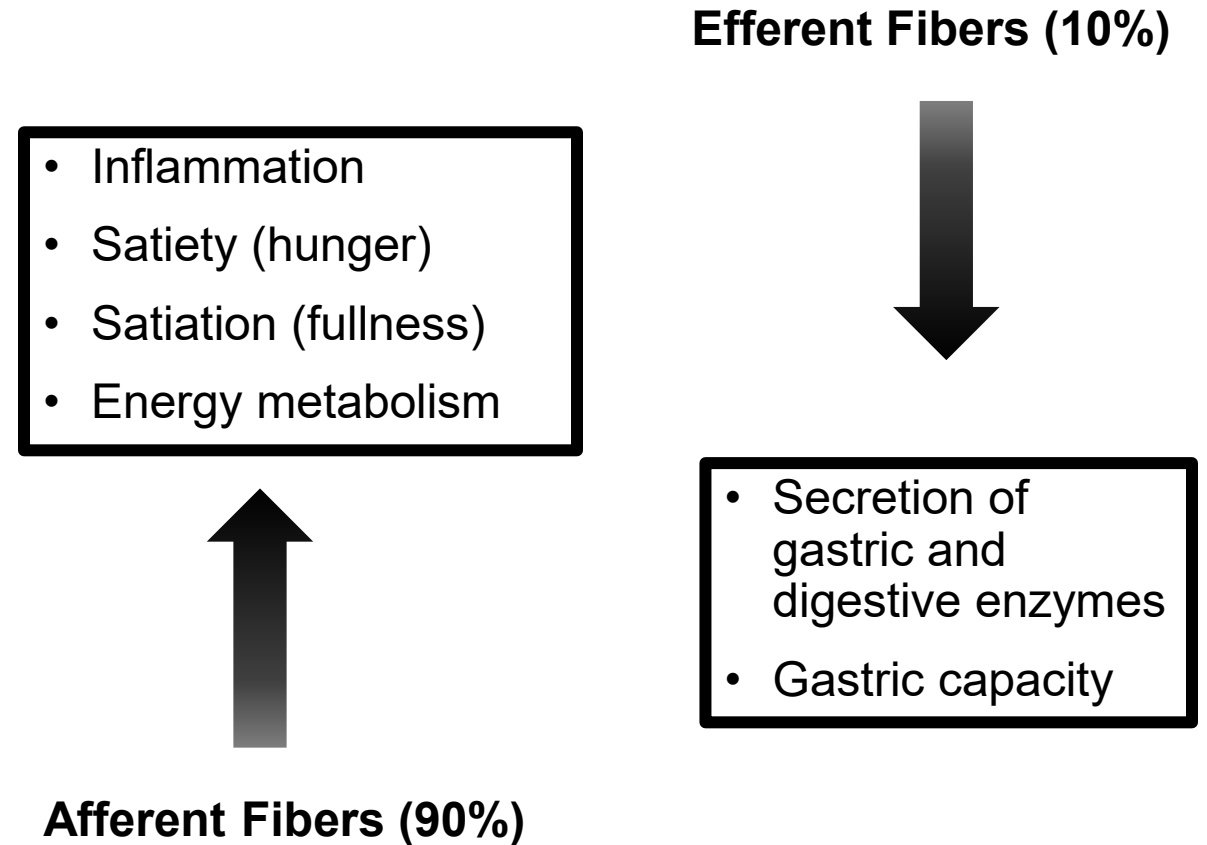
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- Margolis KG, Cryan JF, Mayer EA. The Microbiota-Gut-Brain Axis: From Motility to Mood. *Gastroenterology*. 2021;160(5):1486-1501. doi:10.1053/j.gastro.2020.10.066
- Roth W, Zadeh K, Vekariya R, Ge Y, Mohamadzadeh M. Tryptophan Metabolism and Gut-Brain Homeostasis. *Int J Mol Sci*. 2021;22(6):2973. Published 2021 Mar 15. doi:10.3390/ijms22062973

Long Description: Short chain fatty acids, serotonin, and the microbiota-gut-brain axis

- When short-chain fatty acids stimulate enterochromaffin cells, tryptophan hydroxylase 1 transcription is promoted, allowing tryptophan to be converted into 5 H T P and then serotonin. Gut serotonin signals the C N S and modulates gut motility, neuronal development and differentiation.

Outputs and Outcomes of Afferent and Efferent CNS Signaling



Bidirectional Nature of Gut-Brain Interactions

Roles of the ENS, CNS, Vagus Nerve, and Microbiome

Afferent Signaling

Signals from gut microbiota are transmitted via neurons of the enteric nervous system (ENS) to the vagus nerve, then to the brain.

Chemical Messengers:
SCFAs, Tryptophan, Secondary Bile Acid, Histamine, GABA, DA, NE, 5-HT

Efferent Signaling

The vagus nerve innervates all layers of the gut lining; however, nerve fibers do not cross the gut lining and are not in direct contact with gut microbiota.

Gut microbiota influence the modulation, development, and renewal of ENS neurons.

- Dicks LMT. Our mental health is determined by an intrinsic interplay between the central nervous system, enteric nerves, and gut microbiota. *Int J Mol Sci.* 2023;25(1):38. doi:10.3390/ijms25010038
- Dicks LMT. Gut bacteria and neurotransmitters. *Microorganisms.* 2022;10(9):1838. doi:10.3390/microorganisms10091838
- Chen Y, Xu J, Chen Y. Regulation of neurotransmitters by the gut microbiota and effects on cognition in neurological disorders. *Nutrients.* 2021;13(6):2099. doi:10.3390/nu13062099

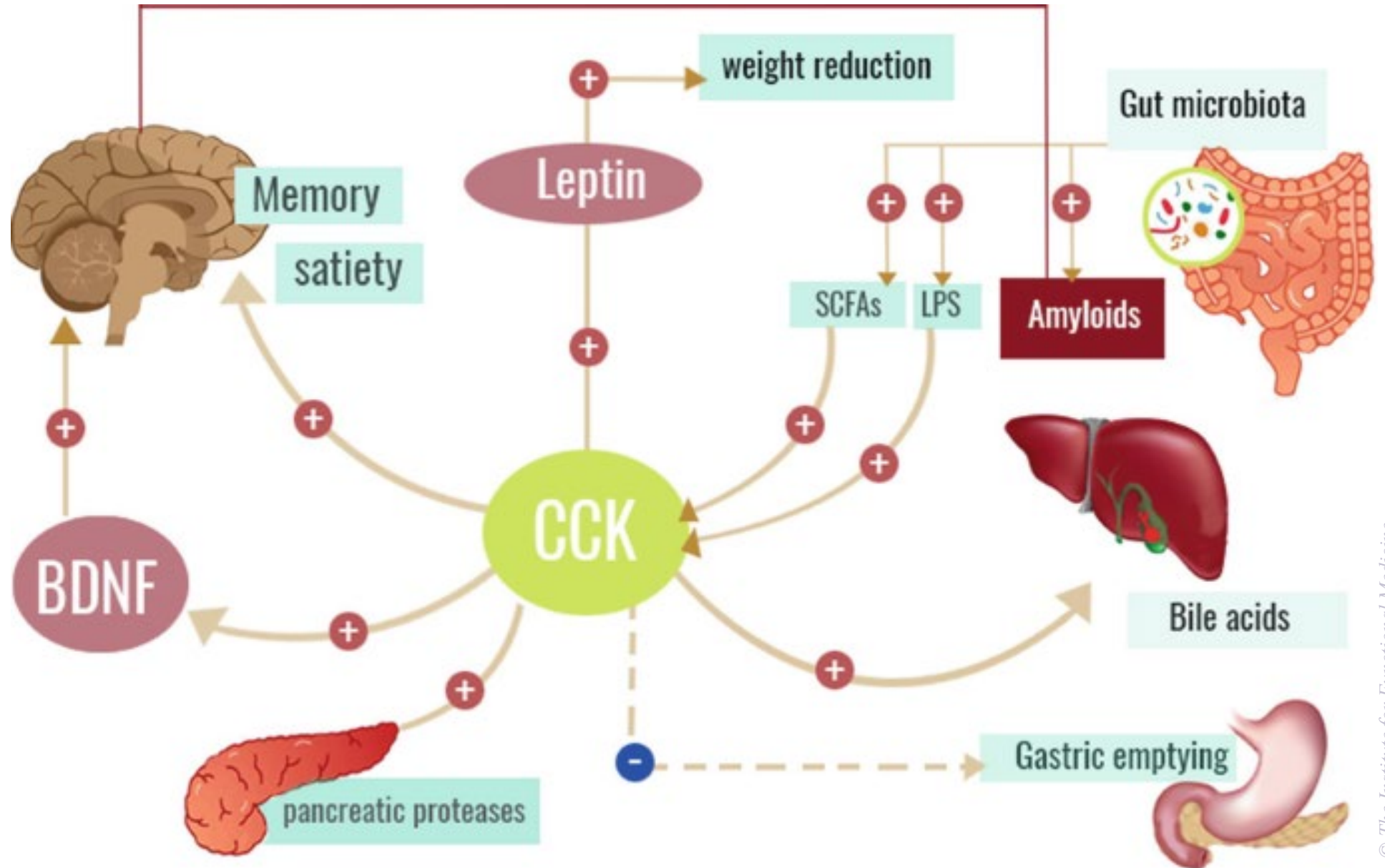
Communication in the Gut-Brain Axis



Enteroendocrine System

- The gut may also be considered an **endocrine organ** due to the presence of enteroendocrine cells (EECs) that secrete hormones and thereby influence appetite, digestion, and metabolism.
- Hormones secreted by EECs include:
 - Cholecystokinin (CCK)
 - 5-hydroxytryptamine (5-HT; serotonin)
 - Glucagon-like peptide-1 (GLP-1)
 - Ghrelin
 - Leptin

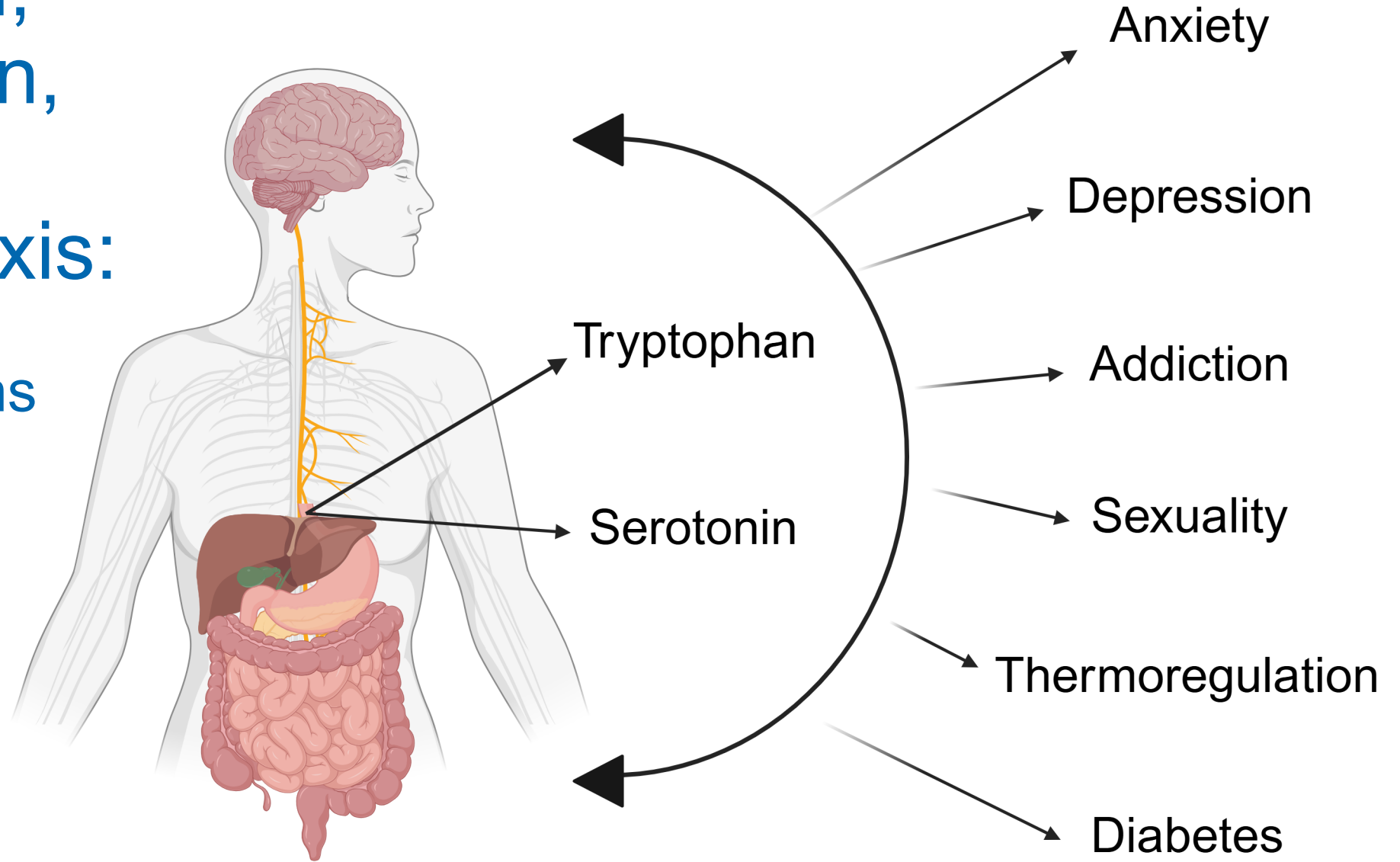
CCK and the Gut-Brain Axis



Long Description: CCK and the Gut-Brain Axis

- Cholecystokinin can be increased by short-chain fatty acids and LPS. CCK stimulates pancreatic proteases, triggers the release of bile acids, promotes satiety signaling through leptin, slows gastric emptying, increases BDNF expression, and may modulate amyloid development.

Serotonin, Tryptophan, and the Gut-Brain Axis: Systemic Manifestations



Long Description: Serotonin, Tryptophan, and the Gut-Brain Axis: Systemic Manifestations

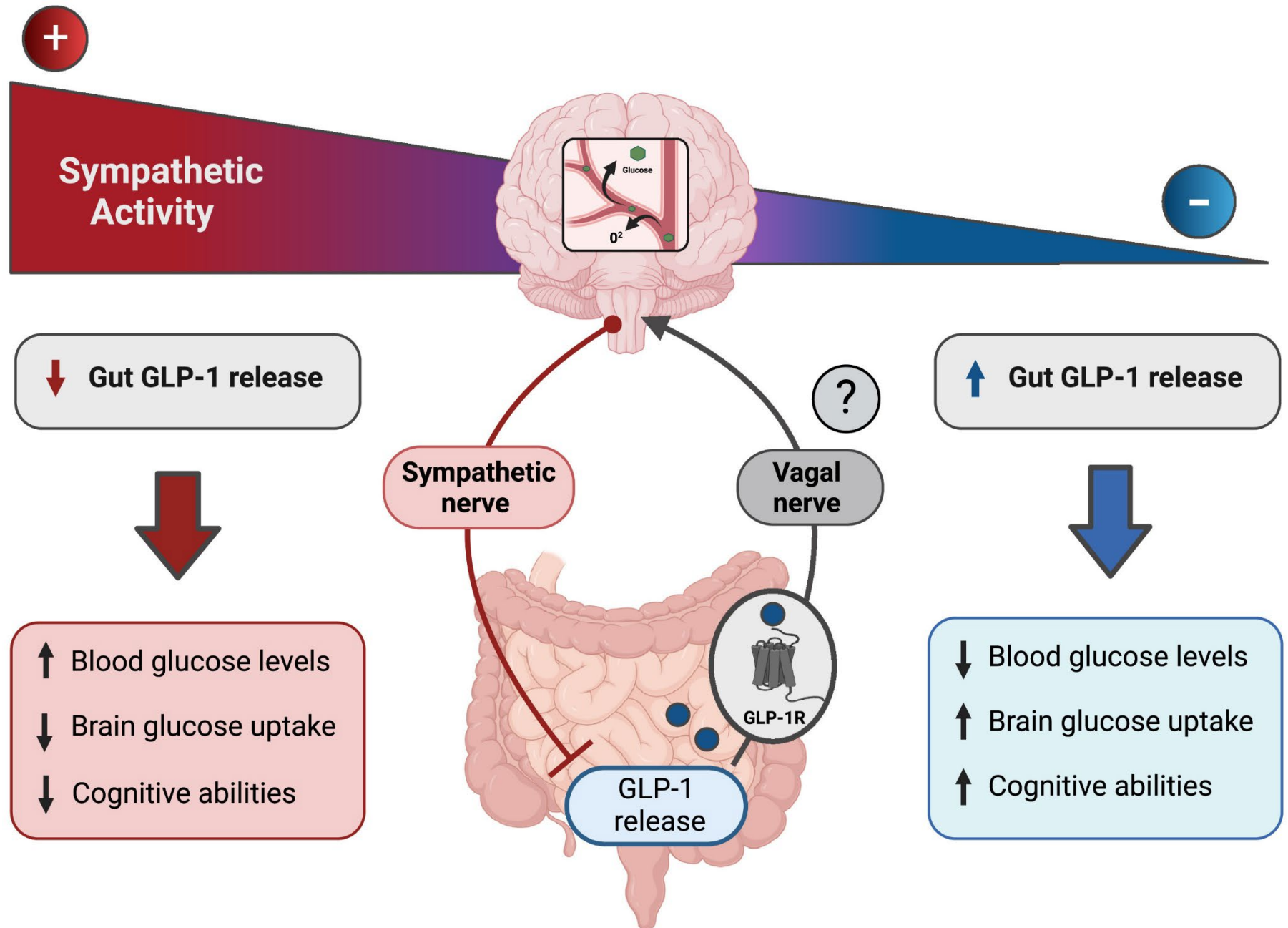
- Serotonin and its precursor, tryptophan, are both produced in the gut. Imbalances have implications in:
 - Anxiety
 - Depression
 - Addiction
 - Sexuality
 - Thermoregulation
 - Diabetes

References

SEROTONIN, TRYPTOPHAN, AND THE GUT-BRAIN AXIS

- Akram N, Faisal Z, Irfan R, et al. Exploring the serotonin-probiotics-gut health axis: a review of current evidence and potential mechanisms. *Food Sci Nutr*. 2023;12(2):694-706. doi:10.1002/fsn3.3826
- Babakhani S, Hosseini F. Gut microbiota: an effective factor in the human brain and behavior. *Neurosci J Shefaye Khatam*. 2019;7(1):106-118. doi:10.29252/shefa.7.1.106
- Kennedy PJ, Cryan JF, Dinan TG, Clarke G. Irritable bowel syndrome: a microbiome-gut-brain axis disorder? *World J Gastroenterol*. 2014;20(39):14105-14125. doi:10.3748/wjg.v20.i39.14105
- O'Mahony SM, Clarke G, Borre YE, Dinan TG, Cryan JF. Serotonin, tryptophan metabolism and the brain-gut-microbiome axis. *Behav Brain Res*. 2015;277:32-48. doi:10.1016/j.bbr.2014.07.027

GLP-1 and the Gut-Brain Axis



Long Description: GLP-1 and the Gut-Brain Axis

- G L P 1 release lowers blood glucose, increases brain glucose uptake, and increases cognition. G L P 1 binds its receptors on afferent vagal nerve fibers, coordinating satiety and metabolic signaling. Under increased sympathetic activity, G L P 1 release is inhibited.

Clinical Focus

- Clinical findings associated with gut-brain axis dysfunction.



Representative Patient Clinical Findings

Gut-Brain Axis Dysfunction

Dysphagia

Obesity/Insulin Resistance

Abdominal Pain

Irritable Bowel Syndrome

Early Satiety

Autoimmunity

Stool Changes

Depression/Anxiety



Clinical Findings of GI Dysfunction

Gut-Brain Axis Dysfunction

Clinical Findings	Pathophysiological Mechanisms
- Dysphagia - Chest pain - Food intolerance - Recurrent aspirations leading to pneumonia	Achalasia is a motility disorder of the lower esophageal sphincter that may be due to a loss of inhibitory neurons within the ENS, resulting in failure of the LES to relax.
- Persistent constipation - Abdominal pain - Nausea, vomiting - Distension	Slow transit constipation is likely due to disruption of both the ENS, ANS, and neuroendocrine system, resulting in decreased innervation of the myenteric plexus, leading to dysmotility of the colon that is unresponsive to dietary changes or laxatives.

See: "References: Clinical Indicators of GI Dysfunction: Dysmotility"

Clinical Findings of GI Dysfunction

Gut-Brain Axis Dysfunction

Clinical Findings	Pathophysiological Mechanisms
• Early satiety • Nausea, vomiting • Postprandial fullness • Distension	Delayed gastric emptying in the absence of an obstruction, as is the case in gastroparesis, is a chronic motility disorder of the stomach that may be due to dysfunctions in both the vagus nerve and ENS that result in abnormal motor neurons and a decrease in enteric neurons.
• Recurrent abdominal pain • Changes in stool frequency and/or form	Changes in sensory and motor functions of the ENS may contribute to IBS symptoms related to dysmotility, such as changes in stool patterns and GI contractions that contribute to abdominal pain.

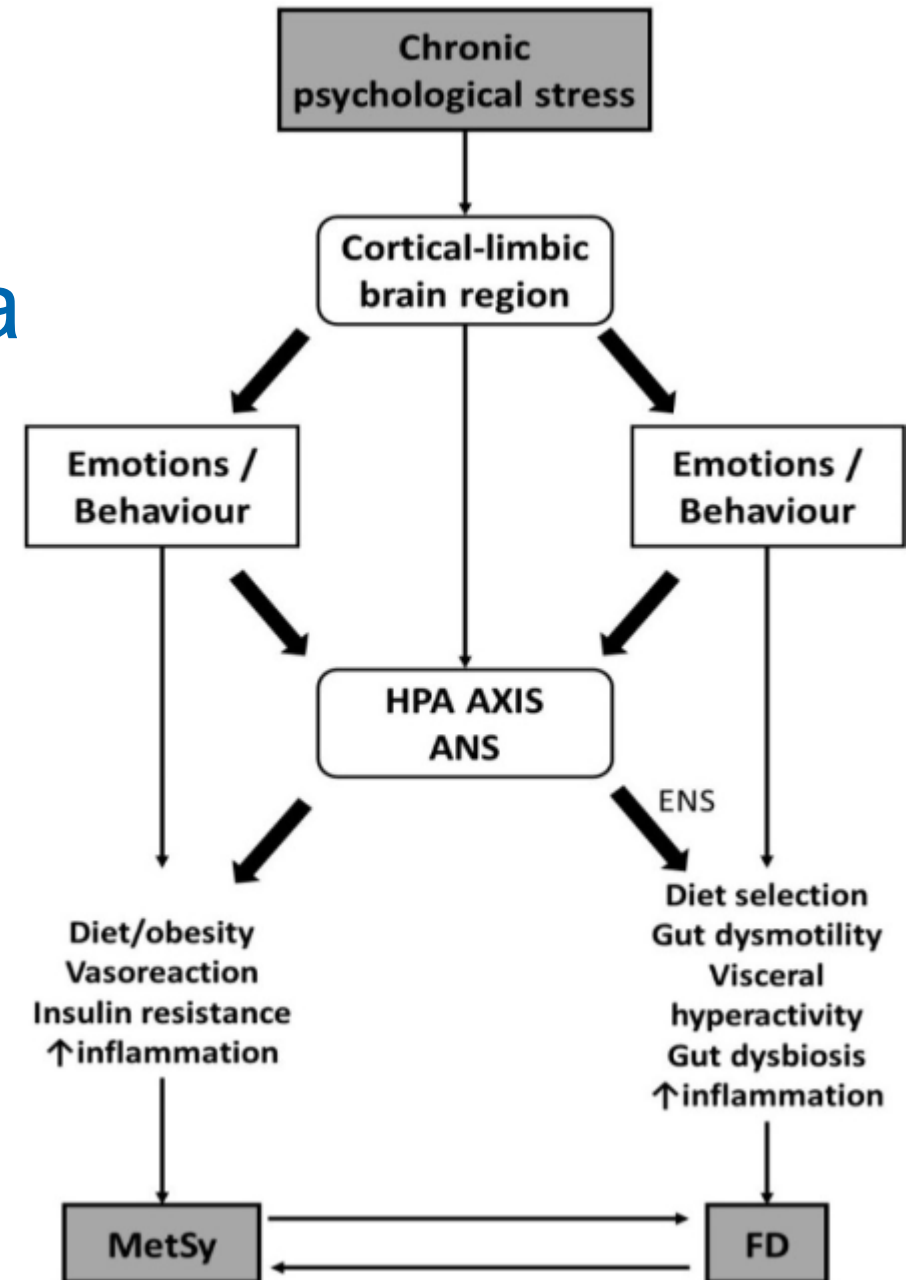
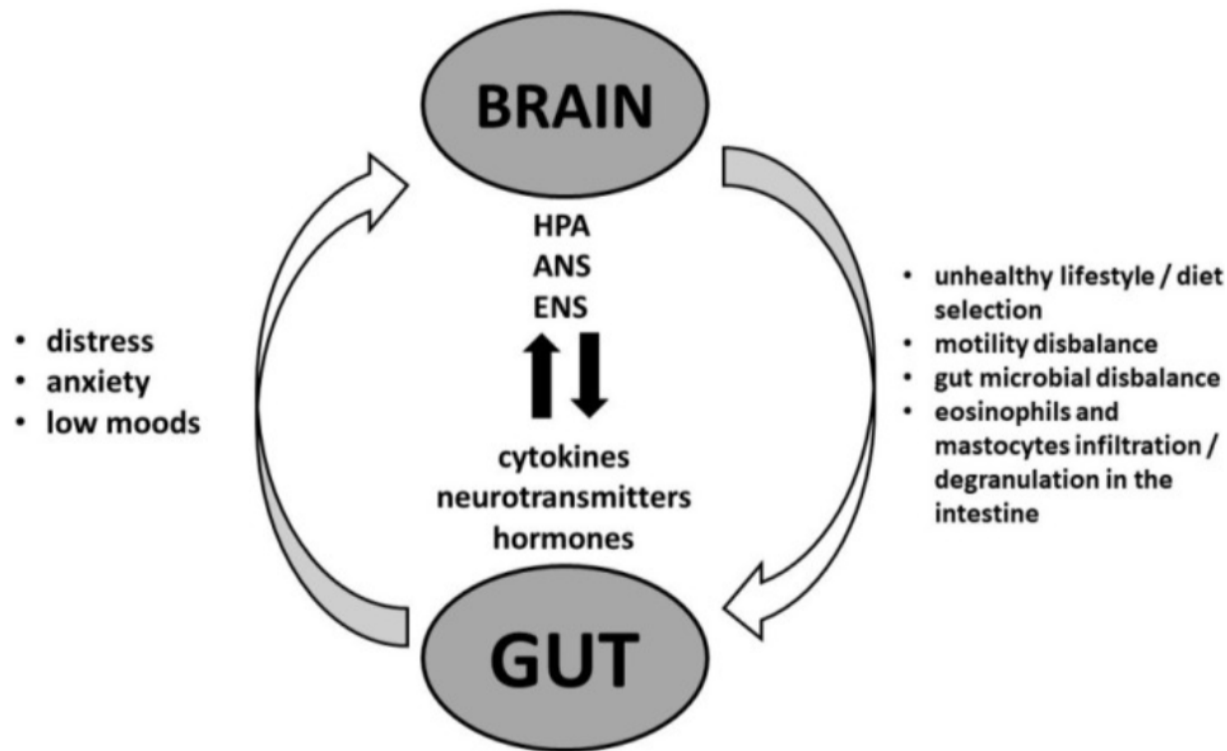
See: "References: Clinical Indicators of GI Dysfunction: Dysmotility"

References

CLINICAL FINDINGS OF GI DYSFUNCTION GUT-BRAIN AXIS DYSFUNCTION

- Fleming MA 2nd, Ehsan L, Moore SR, Levin DE. The enteric nervous system and its emerging role as a therapeutic target. *Gastroenterol Res Pract*. 2020;2020:8024171. doi:10.1155/2020/8024171
- Radocchia G, Marazzato M, Harbi KB, et al. Chronic intestinal pseudo-obstruction: associations with gut microbiota and genes expression of intestinal serotonergic pathway. *BMC Microbiol*. 2024;24(1):48. doi:10.1186/s12866-024-03200-z
- Tobias A, Sadiq NM. *Physiology, Gastrointestinal Nervous Control*. StatPearls Publishing; 2022. Published September 26, 2022. Accessed June 3, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK545268/>

Schematic of Physiology: Gut-Brain Axis – Metabolic Syndrome – Functional Dyspepsia



Long Description: Schematic of Physiology: Gut-Brain Axis, Metabolic Syndrome, Functional Dyspepsia

- Chronic psychological stress disrupts behavior, the H P A axis, and autonomic nervous system function, leading to changes in diet, vascular tone, insulin sensitivity, inflammation, gut microbiota, motility, and visceral sensitivity, all of which contribute to metabolic syndrome and functional dyspepsia.

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Assimilation

- Constipation/bloating
- Heartburn
- Vegetarian, eats quickly

Structural Integrity

- Knee pain

Communication

- Irregular menses
- Elevated fasting insulin

Mental

- Anxious mood
- Stressed

Emotional

Spiritual

Defense & Repair

- Hypothyroid
- Seasonal allergies
- Food sensitivity

Energy

- Poor exercise recovery
- Obesity

Mitigation & Clearance

- Heavy caffeine use
- Constipation

Transport

- Ankle edema
- Cold hands & feet

ENS Dysfunction as Mediator

Long Description: ENS Dysfunction as Mediator

Clinical imbalances and physiological dysfunctions

- Assimilation
 - Constipation, bloating
 - Heartburn
 - Vegetarian, eats quickly
- Structural integrity
 - Knee pain
- Communication
 - Irregular menses
 - Elevated fasting insulin
- Transport
 - Ankle edema
 - Cold hands and feet
- Mitigation and clearance
 - Heavy caffeine use
 - Constipation
- Energy
 - Poor exercise recovery
 - Obesity
- Defense and repair
 - Hypothyroid
 - Seasonal allergies
 - Food sensitivity
- Mental, emotional, and spiritual
 - Anxious mood
 - Stressed

References

ENS DYSFUNCTION AS MEDIATOR

- Nikolla E, Grandberry A, Jamerson D, Flynn CR, Sundaresan S. The enteric neuronal circuitry: a key ignored player in nutrient sensing along the gut-brain axis. *FASEB J.* 2025;39(9):e70586. doi:10.1096/fj.202500220RR
- Chandramowliswaran P, Vijay A, Abraham D, Li G, Mwangi SM, Srinivasan S. Role of sirtuins in modulating neurodegeneration of the enteric nervous system and central nervous system. *Front Neurosci.* 2020;14:614331. doi:10.3389/fnins.2020.614331
- Radocchia G, Marazzato M, Harbi KB, et al. Chronic intestinal pseudo-obstruction: associations with gut microbiota and genes expression of intestinal serotonergic pathway. *BMC Microbiol.* 2024;24(1):48. doi:10.1186/s12866-024-03200-z
- Yan M, Man S, Sun B, et al. Gut liver brain axis in diseases: the implications for therapeutic interventions. *Signal Transduct Target Ther.* 2023;8(1):443. doi:10.1038/s41392-023-01673-4

Clinical Focus

- Stress and other modifiable lifestyle factors as antecedents, triggers, and/or mediators of gut-brain axis dysfunction.
- Primacy of stress-induced dysbiosis as a mechanism that induces dysfunction.



Mediators of ENS Dysfunction

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Inadequate Sleep
Sleep Disturbances

Sedentarism
Overexercise

Processed Foods
Inflammatory Foods

Stress
Gut-Brain Axis Issues

Loneliness
Lack of Support

SLEEP



EXERCISE



NUTRITION



STRESSORS



RELATIONSHIPS



References

MEDIATORS OF ENS DYSFUNCTION

Sleep & Relaxation:

- Vernia F, Di Ruscio M, Ciccone A, et al. Sleep disorders related to nutrition and digestive diseases: a neglected clinical condition. *Int J Med Sci.* 2021;18(3):593-603. doi:10.7150/ijms.45512
- Wang B, Duan R, Duan L. Prevalence of sleep disorder in irritable bowel syndrome: a systematic review with meta-analysis. *Saudi J Gastroenterol.* 2018;24(3):141-150. doi:10.4103/sjg.SJG_603_17
- Yang S, Li SZ, Guo FZ, Zhou DX, Sun XF, Tai JD. Association of sleep duration with chronic constipation among adult men and women: findings from the National Health and Nutrition Examination Survey (2005-2010). *Front Neurol.* 2022;13:903273. doi:10.3389/fneur.2022.903273

Exercise & Movement:

- Cui J, Xie F, Yue H, et al. Physical activity and constipation: a systematic review of cohort studies. *J Glob Health.* 2024;14:04197. doi:10.7189/jogh.14.04197
- Li CQ, Wang YC, Shen SQ, et al. Effects of exercise by type and duration on quality of life in patients with digestive system cancers: a systematic review and network meta-analysis. *J Sport Health Sci.* 2023;12(4):491-500. doi:10.1016/j.jshs.2022.12.008

Nutrition:

- Christensen C, Knudsen A, Arnesen EK, Hatlebakk JG, Sletten IS, Fadnes LT. Diet, food, and nutritional exposures and inflammatory bowel disease or progression of disease: an umbrella review. *Adv Nutr.* 2024;15(5):100219. doi:10.1016/j.advnut.2024.100219
- Magro DO, Sasaki LY, Chebli JMF. Interaction between diet and genetics in patients with inflammatory bowel disease. *World J Gastroenterol.* 2024;30(12):1644-1650. doi:10.3748/wjg.v30.i12.1644
- Rostami A, White K, Rostami K. Pro and anti-inflammatory diets as strong epigenetic factors in inflammatory bowel disease. *World J Gastroenterol.* 2024;30(27):3284-3289. doi:10.3748/wjg.v30.i27.3284

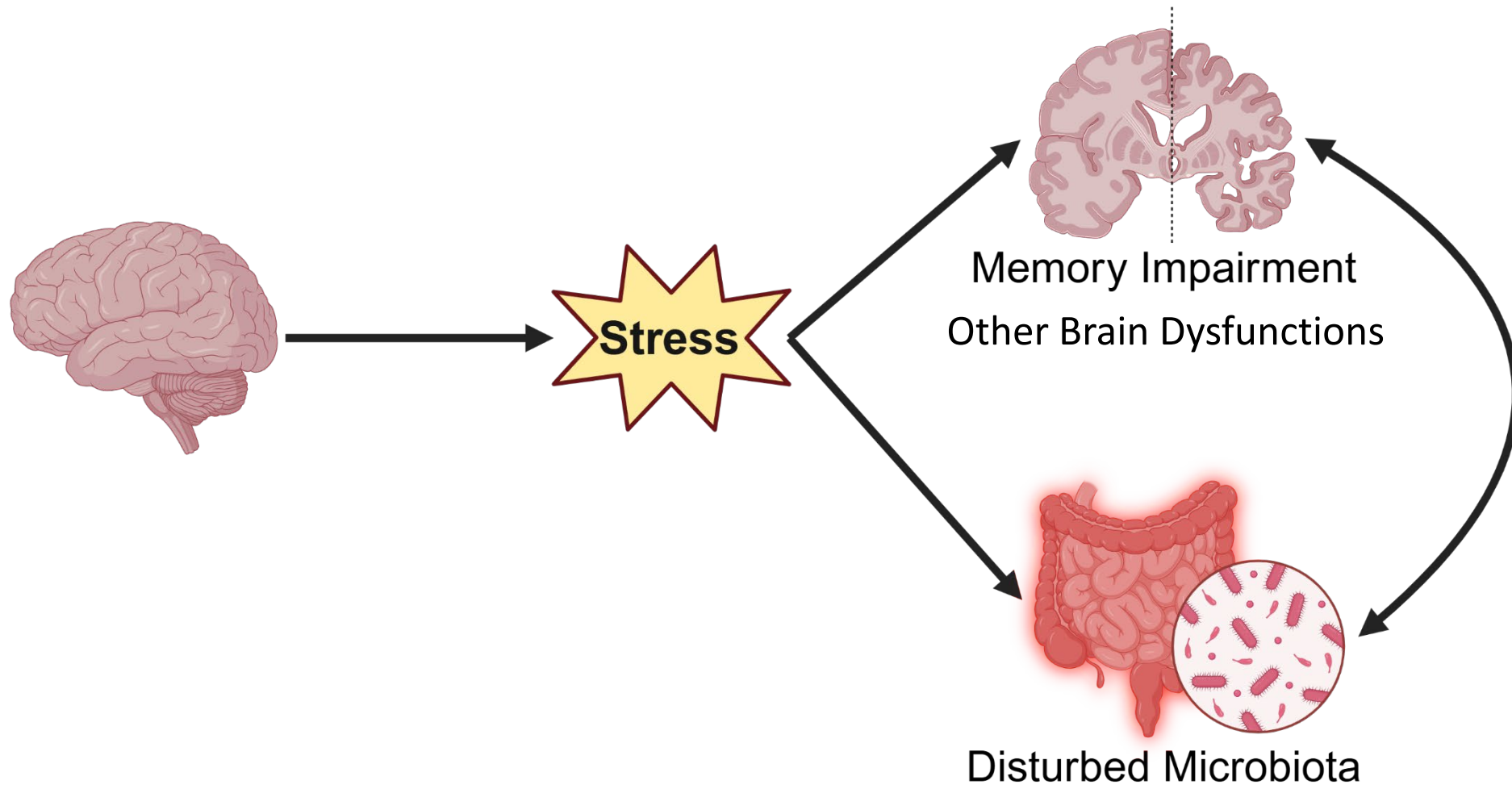
Stress:

- Glynn H, Möller SP, Wilding H, Apputhurai P, Moore G, Knowles SR. Prevalence and impact of post-traumatic stress disorder in gastrointestinal conditions: a systematic review. *Dig Dis Sci.* 2021;66(12):4109-4119. doi:10.1007/s10620-020-06798-y

Relationships:

- Pourriyahi H, Yazdanpanah N, Saghaizadeh A, Rezaei N. Loneliness: an immunometabolic syndrome. *Int J Environ Res Public Health.* 2021;18(22):12162. doi:10.3390/ijerph182212162
- Nguyen TT, Zhang X, Wu TC, et al. Association of loneliness and wisdom with gut microbial diversity and composition: an exploratory study. *Front Psychiatry.* 2021;12:648475. doi:10.3389/fpsy.2021.648475
- Luchetti M, Aschwanden D, Sesker AA, et al. A Meta-analysis of Loneliness and Risk of Dementia using Longitudinal Data from >600,000 Individuals. *Nat Ment Health.* 2024;2(11):1350-1361. doi:10.1038/s44220-024-00328-9

Representative Example of Stress/Dysbiosis Consequences: Memory Impairment



Long Description: Example of Stress/Dysbiosis

Consequences: Memory Impairment

- Stress disrupts memory, brain function, and gut microbiota, each of which can further dysregulate the others in a bidirectional relationship.

References

EXAMPLES OF STRESS/DYSBIOSIS CONSEQUENCES: MEMORY IMPAIRMENT

- An J, Liu Y, Wang Y, et al. The role of intestinal mucosal barrier in autoimmune disease: a potential target. *Front Immunol*. 2022;13:871713. doi:10.3389/fimmu.2022.871713
- Tokuhara D, Kurashima Y, Kamioka M, Nakayama T, Ernst P, Kiyono H. A comprehensive understanding of the gut mucosal immune system in allergic inflammation. *Allergol Int*. 2019;68(1):17-25. doi:10.1016/j.alit.2018.09.004
- Cristofori F, Dargenio VN, Dargenio C, Miniello VL, Barone M, Francavilla R. Anti-inflammatory and immunomodulatory effects of probiotics in gut inflammation: a door to the body. *Front Immunol*. 2021;12:578386. doi:10.3389/fimmu.2021.578386
- Childs CE, Calder PC, Miles EA. Diet and immune function. *Nutrients*. 2019;11(8):1933. doi:10.3390/nu11081933
- Wallensten J, Ljunggren G, Nager A, et al. Stress, depression, and risk of dementia - a cohort study in the total population between 18 and 65 years old in Region Stockholm. *Alzheimers Res Ther*. 2023;15(1):161. doi:10.1186/s13195-023-01308-4

Early Life Stress (SDOH and ACEs) as Antecedents of IBS

- All areas of DIGIN can be affected in the context of IBS.
- Maternal stress levels and genetic predispositions impact brain-gut-microbiome (BGM) development.
- Mode of delivery, early adverse life events, and early nutrition also result in brain-gut phenotypes that influence the response to the exposome in adulthood.

ENS/Gut-Brain Axis as Mediator of GI Dysfunctions

Root Causes and Mechanisms of Action

ENS-Induced Effects	GI Dysfunction (DIGIN Dysfunction)	Root Causes and Mechanisms of Action
Decreased appetite Nutrient imbalances and/or deficiencies	Digestion/Absorption Gut Microbiota	Areas of the brain that are responsible for feeling pleasure and motivation with eating (ventral tegmental area – VTA), and processing stress-related emotions (amygdala) can both decrease the desire to eat. Nutrition can also impact stress response (i.e., bidirectional relationship).
Dysmotility - Constipation - Gastroparesis - Incontinence - Evacuation ailments	Digestion/Absorption Gut Microbiota	ENS acts independently of VN in SI and LI via its own sensory and motor neurons that regulate muscle activity, gut wall motility, and mucosal barrier integrity. - Increased TGFβ2 activity result in immature ENS neurons that are unable to coordinate normal smooth muscle activity. - CRH also plays into increased movement of terminal sections of GI tract and decreased movements of proximal sections of GI tract.

See: "References: ENS/Gut-Brain Axis as Mediator of GI Dysfunctions"

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ENS-Induced Effects	GI Dysfunction (DIGIN Dysfunction)	Root Causes and Mechanisms of Action
Impairment of gastric, pancreatic, gallbladder, and/or brush border function	Maldigestion/Malabsorption	ENS acts independently of VN in SI and LI, via its own sensory and motor neurons that regulate fluid secretion, blood flow, and mucosal barrier integrity.
ENS-induced Inflammation	Intestinal Permeability Gut Microbiome/Dysbiosis Immune Dysfunction/Gut Inflammation	ENS influenced to produce the pro-inflammatory signal/cytokine (CSF-1) to recruit immune cells (monocytes → macrophages), which can result in inflammation (review 301D) and intestinal permeability.

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References

ENS/GUT-BRAIN AXIS AS MEDIATOR OF GI DYSFUNCTIONS

- Chaudhry TS, Senapati SG, Gadam S, et al. The impact of microbiota on the gut-brain axis: examining the complex interplay and implications. *J Clin Med*. 2023;12(16):5231. doi:10.3390/jcm12165231
- Dicks LMT. Gut bacteria and neurotransmitters. *Microorganisms*. 2022;10(9):1838. doi:10.3390/microorganisms10091838
- Fleming MA 2nd, Ehsan L, Moore SR, Levin DE. The enteric nervous system and its emerging role as a therapeutic target. *Gastroenterol Res Pract*. 2020;2020:8024171. doi:10.1155/2020/8024171
- Schneider KM, Blank N, Thaiss CA. From mental strain to gut pain: a brain-gut pathway transducing psychological stress to intestinal inflammation. *Clin Transl Med*. 2023;13(10):e1458. doi:10.1002/ctm2.1458
- Yaribeygi H, Panahi Y, Sahraei H, Johnston TP, Sahebkar A. The impact of stress on body function: a review. *EXCLI J*. 2017;16:1057-1072. doi:10.17179/excli2017-480

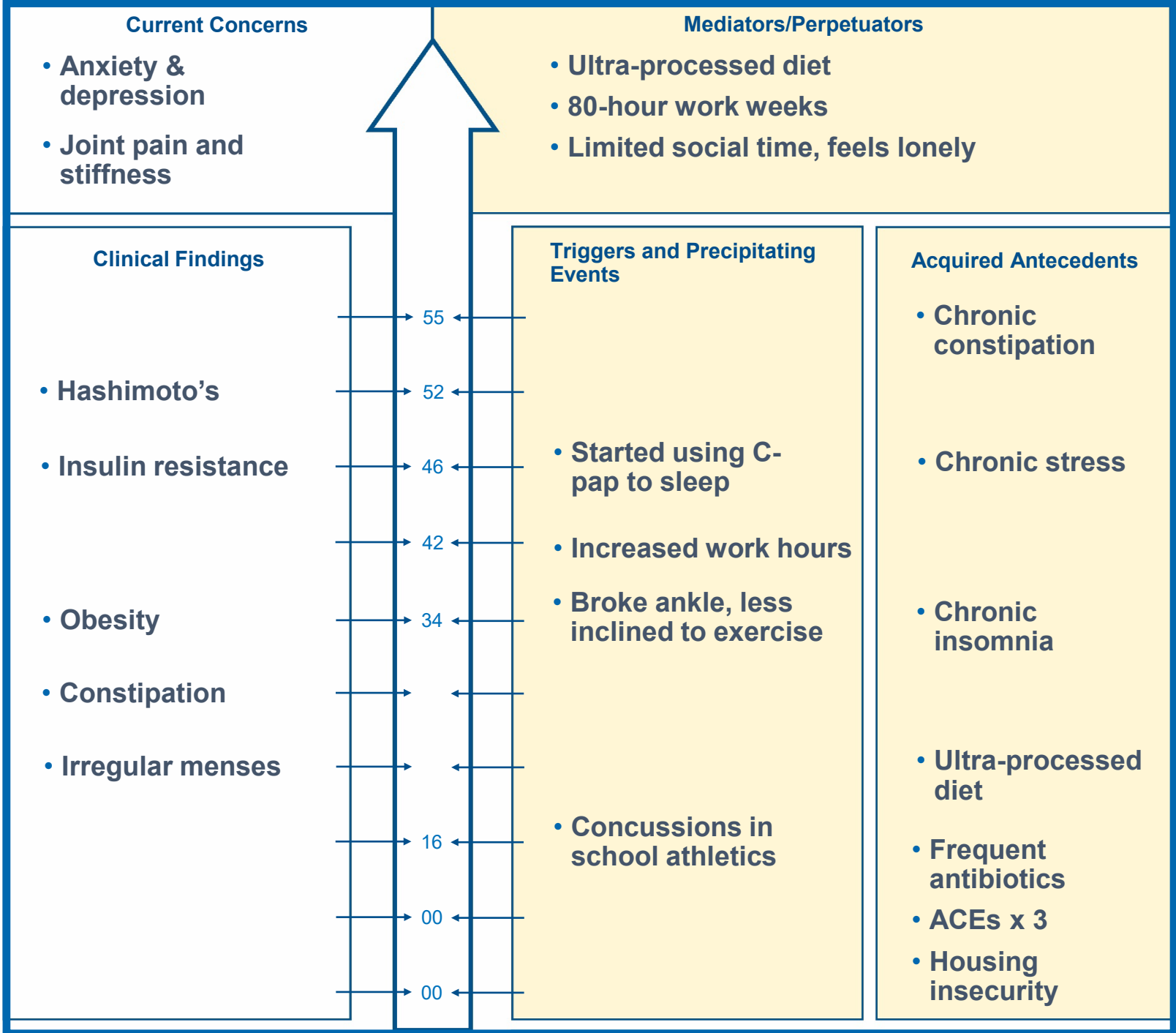
Case Vignette

Antecedents, Triggers, Mediators

- Sleep apnea
- Sedentary lifestyle
- Diet high in ultra-processed and inflammatory foods
- Works 80 hours per week
- Not much time for a social life – feels lonely often

Extraintestinal Manifestations

- Anxiety and depression
- Joint pain and stiffness
- Autoimmunity (Hashimoto's)
- Insulin resistance



Gut-Brain Axis on the Timeline

A clinical case example

Long Description: Gut-Brain Axis on the Timeline

- Current Concerns
 - Anxiety and depression
 - Joint pain and stiffness
- Clinical Findings
 - Hashimoto's
 - Insulin resistance
 - Hypertension
 - Obesity
 - Constipation
 - Irregular menses
- Mediators/Perpetuators
 - Ultra-processed diet
 - 80-hour work weeks
 - Limited social time, feels lonely
- Acquired Antecedents
 - Chronic constipation
 - Chronic stress
 - Chronic insomnia
 - Ultra-processed diet
 - Frequent antibiotics
 - 3 adverse childhood events
 - Housing insecurity
- Triggers and Precipitating Events
 - Started using C-pap to sleep
 - Increased work hours
 - Broke ankle, less inclined to exercise
 - Concussions in school athletics

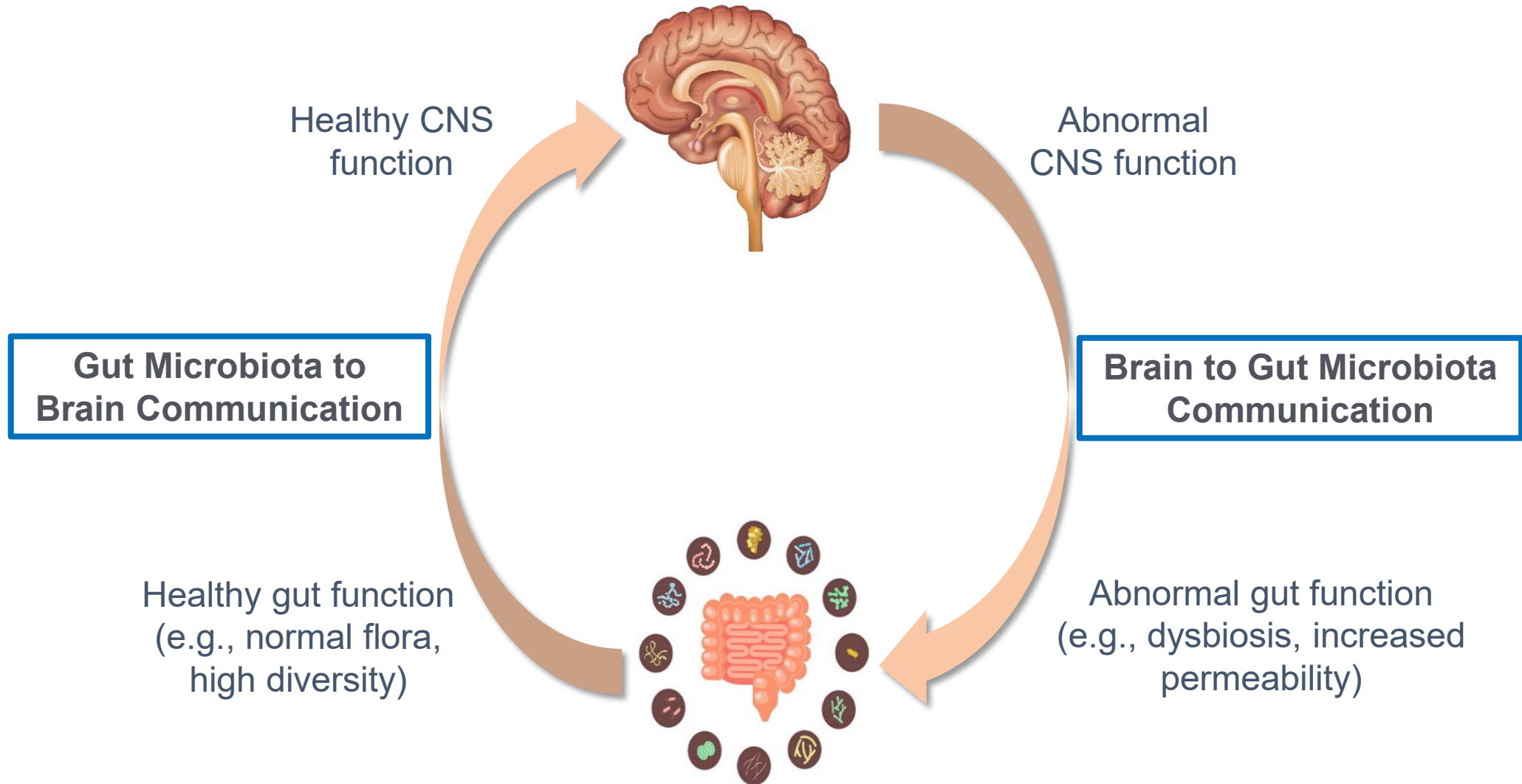
Clinical Focus

- Summary, conclusion, and key clinical takeaways.



Gut-Brain Axis

Affects Behavior, Cognition, Emotion, Inflammatory Cells, and Mediators



- Al-Asmakh M, Anuar F, Zadjali F, Rafter J, Pettersson S. Gut microbial communities modulating brain development and function. *Gut microbes*. 2012;3(40):366-373. doi:10.4161/gmic.21287
- Cryan JF, Dinan TG. Mind-altering microorganisms: the impact of the gut microbiota on brain and behaviour. *Nat Rev Neurosci*. 2012;13(10):701-712. doi:10.1038/nrn3346
- Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Gut-Brain Axis

- A bidirectional relationship exists between the gut microbiota and the central nervous system. Healthy gut function supports healthy C N S function, while C N S dysfunction can lead to abnormal gut function.

Clinical Findings of GI Dysfunction

Gut-Brain Axis Dysfunction

Clinical Findings	Pathophysiological Mechanisms
- Dysphagia - Chest pain - Food intolerance - Recurrent aspirations leading to pneumonia	Achalasia is a motility disorder of the lower esophageal sphincter that may be due to a loss of inhibitory neurons within the ENS, resulting in failure of the LES to relax.
- Persistent constipation - Abdominal pain - Nausea, vomiting - Distension	Slow transit constipation is likely due to disruption of both the ENS, ANS, and neuroendocrine system, resulting in decreased innervation of the myenteric plexus, leading to dysmotility of the colon that is unresponsive to dietary changes or laxatives.

See: "References: Clinical Indicators of GI Dysfunction: Dysmotility"

Clinical Findings of GI Dysfunction

Gut-Brain Axis Dysfunction

Clinical Findings	Pathophysiological Mechanisms
• Early satiety • Nausea, vomiting • Postprandial fullness • Distension	Delayed gastric emptying in the absence of an obstruction, as is the case in gastroparesis, is a chronic motility disorder of the stomach that may be due to dysfunctions in both the vagus nerve and ENS that result in abnormal motor neurons and a decrease in enteric neurons.
• Recurrent abdominal pain • Changes in stool frequency and/or form	Changes in sensory and motor functions of the ENS may contribute to IBS symptoms related to dysmotility, such as changes in stool patterns and GI contractions that contribute to abdominal pain.

See: "References: Clinical Indicators of GI Dysfunction: Dysmotility"

ENS/Gut-Brain Axis as Mediator of GI Dysfunctions

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See: "References: ENS/Gut-Brain Axis as Mediator of GI Dysfunctions"

Mediators of ENS Dysfunction

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

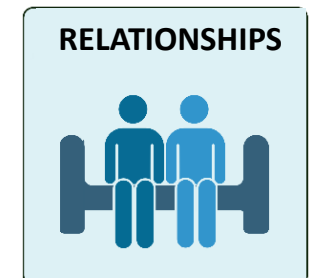
Inadequate Sleep
Sleep Disturbances

Sedentarism
Overexercise

Processed Foods
Inflammatory Foods

Stress
Gut-Brain Axis Issues

Loneliness
Lack of Support



Enteric Nervous System Dysfunction:

Manifestations – Mediators – Mitigators

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE